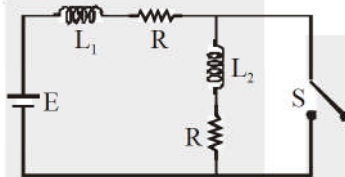
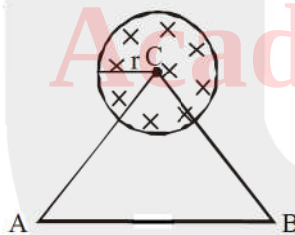


[SINGLE CORRECT CHOICE TYPE]

- Q.1 In a uniform magnetic field of 10^{-5} T in free space, the energy density is u . The electric field which will produce the same energy density in free space is
(A) 10^5 V/m (B) 3×10^3 V/m (C) 10 V/m (D) 9×10^{-3} V/m
- Q.2 The switch S shown in the figure is closed for $t < 0$ and is opened at $t = 0$. When currents through L_1 and L_2 are equal, their common value is

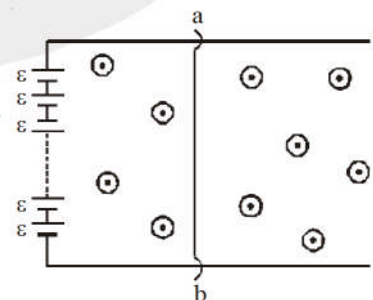


- (A) $\frac{E}{R}$ (B) $\frac{E(L_2 + L_1)}{RL_1}$ (C) $\frac{EL_1}{R(L_1 + L_2)}$ (D) $\frac{E(L_1 + L_2)}{R L_2}$
- Q.3 A uniform magnetic field of intensity $B = B_0 \sin(\omega t)$ directed into the plane of the paper exists in the cylindrical region of radius r . A loop of resistance $R = 5\Omega$ is folded in the form of an equilateral triangle of side length $2r$ is placed as shown in the figure. The maximum potential drop in the wire AB is



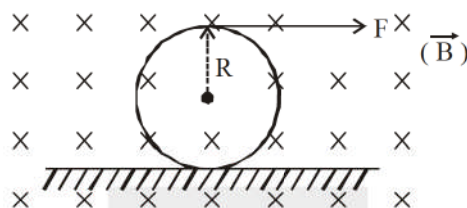
- (A) $\frac{\pi r^2 B_0 \omega}{6}$ volt (B) $\frac{\pi r^2 B_0 \omega}{3}$ volt (C) $\frac{\pi r^2 B_0 \omega}{2}$ volt (D) $\frac{\pi r^2 B_0 \omega}{9}$ volt
- Q.4 A conducting loop of radius R is present in a uniform magnetic field B perpendicular to the plane of ring. If radius R varies as a function of time t as $R = R_0 + t^2$. The emf induced in the loop is
(A) $2\pi Bt(R_0 + t^2)$ clockwise (B) $4\pi Bt^2(t^2 + R_0)$ anticlockwise
(C) $4\pi Bt(t^2 + R_0)$ anticlockwise (D) $4\pi B(t^2 + R_0)$ clockwise.

- Q.5 In figure n identical battery having emf ϵ having negligible internal resistances with one battery having reverse polarity is connected to a smooth pair of thick metallic rails. A wire ab of length ℓ and resistance r can slide smoothly on the rails. There exists a uniform magnetic field B perpendicular to the plane of frame. The wire is given a small velocity v towards right. The current in wire ab at the very instant will be [3]

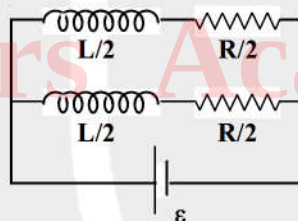


- (A) $\frac{(n-2)\epsilon - Bv\ell}{r}$ (B) $\frac{n\epsilon - Bv\ell}{r}$
(C) $\frac{(n-1)\epsilon - Bv\ell}{r}$ (D) $\frac{(n-2)\epsilon + Bv\ell}{r}$

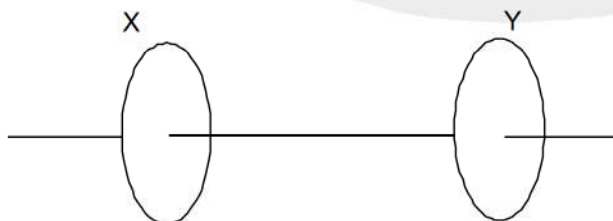
- Q.6 A force F is applied at the top of a uniform conducting ring of radius R and mass m which is kept at rest on a horizontal ground surface. A uniform magnetic field B is acting perpendicular to the plane of ring as shown in figure. Then potential difference between the top and bottom ends of the ring after time t is [3]



- (A) zero (B) $\frac{FBRt}{m}$ (C) $\frac{2FBRt}{m}$ (D) $\frac{4FBRt}{m}$
- Q.7 A solenoid has 2000 turns wound over a length of 0.30 m. The area of its cross section is $1.2 \times 10^{-3} \text{ m}^2$. Around its central section a coil of 300 turns is wound. If an initial current of 2 A in the solenoid is reversed in 0.25 s, the emf induced in the coil is equal to
 (A) $6 \times 10^{-4} \text{ V}$ (B) $4.8 \times 10^{-2} \text{ V}$ (C) $6 \times 10^{-2} \text{ V}$ (D) 48 kV
- Q.8 A $50/\pi$ Hz a.c. current of crest value 1 A flows through the primary of a transformer. If the mutual inductance between the primary and secondary be 1.5 H, the crest voltage induced in the secondary is
 (A) 75 V (B) 150 V (C) 225 V (D) 300 V
- Q.9 For the shown network, time constant of the circuit across the terminals of the battery is equal to



- (A) L/R (B) $L/2R$ (C) $2L/R$ (D) none of these
- Q.10 The self-inductance of two coils is 9 henry and 4 henry respectively. Find the maximum mutual inductance produced by these coils
 (A) 10 Henry (B) $\frac{9}{4}$ Henry (C) 6 Henry (D) $\frac{4}{9}$ Henry
- Q.11 The two loops shown in the figure have their planes parallel to each other. Current flows in the loop X. The two coils will repel each other if the current in the loop X is

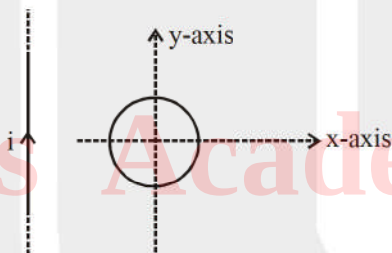


- (A) increasing (B) decreasing (C) constant (D) None of these

- Q.12 In an a.c. circuit V and I are given by $V = 50 \sin 50t$ volt and $I = 100 \sin(50t + \pi/3)$ mA. The power dissipated in the circuit
 (A) 2.5 watt (B) 1.25 watt (C) 5.0 watt (D) 500 watt
- Q.13 In a series resonant circuit, the a.c. voltage across resistance R , inductance L and capacitance C are 5V, 10V and 10V, respectively. The a.c. voltage applied to the circuit will be
 (A) 20V (B) 10V (C) 5V (D) 25V
- Q.14 An LCR circuit contains resistance of 100Ω and a supply of 200V at 300rad/s angular frequency. If only capacitance is taken out from the circuit and the rest of the circuit is connected, current lags behind the voltage by 60° . If on the other hand only inductor is taken out, the current leads by 60° with the applied voltage. The current flowing in the circuit is
 (A) 1A (B) 1.5A (C) 2A (D) 2.5A
- Q.15 In LCR circuit the capacitance is changed from C to $4C$. For the same resonant frequency, the inductance should be changed from L to
 (A) $2L$ (B) $L/2$ (C) $L/4$ (D) $4L$

[MATRIX TYPE]

- Q.16 A conducting circular rigid loop near a long straight current carrying wire is shown. Match the following table :



Column I

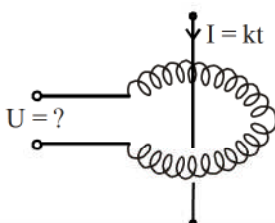
- (A) If current is increased
 (B) If current is decreased
 (C) If loop is moved away from the wire maintaining constant current in the straight wire.

Column II

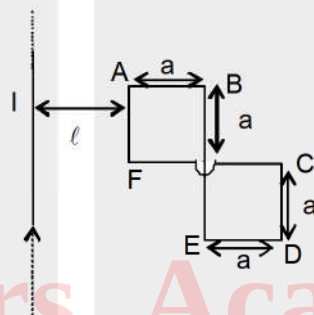
- (P) Induced current in the loop is clockwise
 (Q) Induced current in the loop is anticlockwise
 (R) Wire will attract the loop and there will be no torque about y-axis
 (S) Wire will repel the loop and there will be no torque about y-axis

[SUBJECTIVE TYPE]

- Q.17 To a circuit of 1Ω resistance and 0.01 henry inductance, is connected a 200Volt line of frequency 50 cycles/ second. Calculate the reactance, the impedance and the current in the circuit and also the lag in phase between alternating voltage and current.
- Q.18 In figure a toroid of radius 4m is made of coil of 2000 turns and cross-section area 1 cm^2 . What is the voltage across the terminals of the coil if a long straight wire along the axis carries a current $I = 8t$, where t is time.



- Q.19 A current of 4A flows in a coil when connected to a 12 V d.c. source. If the same coils is connected to a 12 V, 50 rad/s, a.c. source, a current of 2.4 A flows in the circuit. Determine the inductance of coil. Also find the power developed in the circuit if a 2500 μF condenser is connected in series with the coil.
- Q.20 A circuit draws a power of 0.4 KW from a source of 200 volt, $\frac{24}{\pi}$ Hz. The power factor of the circuit is 0.8 and the current lags behind the source voltage.
 (a) What single element inductance or capacitance connected to the circuit and its value.
 (b) In order to make the power factor 1.0 what electrical element inductance or capacitance and its value to be connected in the circuit?
- Q.21 In the figure shown there is a conducting loop ABCDEF of resistance per unit length placed near by a long straight current carrying wire. The dimensions are shown in the figure. The long wire lies in the plane of the loop. The current in the long wire varies as $I = I_0 t$. Find
 (a) the mutual inductance of the wire and the loop
 (b) the heat produced in the loop in time t .



ANSWER KEY

Q.1	B	Q.2	C	Q.3	D	Q.4	C	Q.5	D	Q.6	C	Q.7	B
Q.8	B	Q.9	A	Q.10	C	Q.11	A	Q.12	B	Q.13	C	Q.14	C
Q.15	C	Q.16	(A) Q,S (B) P,R (C) P,R				Q.17	3.14, 3.3, 200/3.3 A, $\tan^{-1}\pi$					
Q.18	8×10^{-8} volts			Q.19	0.08 H, 17.28 W			Q.20	(i) 1 H, (ii) 434 μF				

Q.21 $M = \frac{\mu_0 a}{2\pi} \ln \frac{(a+l)^2}{\ell(2a+\ell)}$, $\varepsilon = -MI_0$, Heat produced $= \frac{\varepsilon^2}{R} t$

